

Uniform and Pointwise Convergence for Sequences of Functions on $[0,1]$

We look at two notions of convergence for sequences of functions. The stronger one—uniform convergence—preserves continuity of functions.

What I assume students have seen

- Sequences of real numbers and their convergence

(Reminder: we say that $(a_n)_{n \in \mathbb{N}}$ converges to L if for all $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ s.t. $|a_n - L| < \epsilon$ for all $n \geq n_0$)

- The definition of a sequence of functions and some examples.

(Definition)

Let $D \subseteq \mathbb{R}$ and, for each n in $\mathbb{N}$, let f_n be a function from D into $\mathbb{R}$. Then, $(f_n)_{n \in \mathbb{N}}$ is a sequence of functions on D . For each $x \in D$, the sequence $(f_n(x))_{n \in \mathbb{N}}$ is a sequence of real numbers.

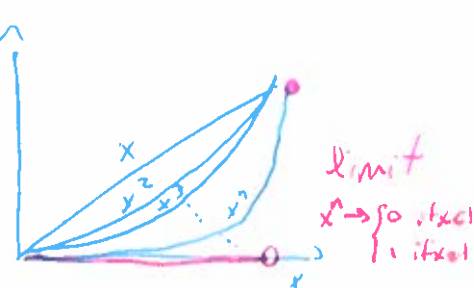
Remark: Two types of variables:
 $n \in \mathbb{N}$, $x \in \mathbb{R}$

Examples

$$\left(\frac{x}{n}\right)_{n \in \mathbb{N}}$$



$$(x^n)_{n \in \mathbb{N}}$$



(2)

Convergence

Definition

If, for every $x \in D$, $\underbrace{(f_n(x))}_{\substack{\text{real number} \\ \text{for fixed } x}}_{n \in \mathbb{N}}$ converges, then the sequence of functions $(f_n(x))_{n \in \mathbb{N}}$ converges pointwise.

Its limit, $F(x) = \lim_{n \rightarrow \infty} f_n(x)$, is the limit function.

Remark: The limit function satisfies:

"For all $\epsilon > 0$ and for all $x \in D$, there exists $n_0 \in \mathbb{N}$ s.t.

$$|f_n(x) - F(x)| < \epsilon \quad \forall n \geq n_0. "$$

Examples

$(x/n)_{n \in \mathbb{N}}$.

For a fixed $x \in [0, 1]$, $0 \leq \frac{x}{n} \leq \frac{1}{n}$.

By the squeeze theorem, $\frac{x}{n} \rightarrow 0$ as $n \rightarrow \infty$.

Examples

$(x^n)_{n \in \mathbb{N}}$

• For $x \in [0, 1)$, $x^n \rightarrow 0$ as $n \rightarrow \infty$

• For $x = 1$, $x^n = 1^n \rightarrow 1$ as $n \rightarrow \infty$

$\{x^n\}_{n \in \mathbb{N}}$ converges pointwise to $F(x) = \begin{cases} 0 & \text{if } x \in [0, 1) \\ 1 & \text{if } x = 1 \end{cases}$

(3)

Problem: • x^n is continuous on $[0,1]$ for all $n \in \mathbb{N}$

• $\lim_{n \rightarrow \infty} x^n$ is not continuous on $[0,1]$

Pointwise convergence does not preserve continuity...

Uniform convergence

Definition

Given a sequence of functions $(f_n)_{n \in \mathbb{N}}$ on D and a function F on D , $(f_n)_{n \in \mathbb{N}}$ converges uniformly to F if, for all $\epsilon > 0$, there exists $n_0 \in \mathbb{N}$ such that

$$|f_n(x) - F(x)| < \epsilon \quad \text{for all } n > n_0 \text{ and all } x \in D.$$

Remark: n_0 depends only on ϵ , not on x .

Why we care? Uniform convergence preserves continuity (Theorem, coming later).

Example

$(\frac{x}{n})_{n \in \mathbb{N}}$ converges uniformly to 0 on $D = [0,1]$

Proof: Let $\epsilon > 0$. We know $|\frac{x}{n}| < \frac{1}{n}$. Let $n_0 \in \mathbb{N}$ be s.t. $\frac{1}{n_0} < \epsilon$ (exists by Archimedean Principle). Then, if $x \in [0,1]$ and $n \geq n_0$,

$$|\frac{x}{n} - 0| \leq \frac{1}{n} \leq \frac{1}{n_0} < \epsilon.$$

So $(\frac{x}{n})_{n \in \mathbb{N}}$ converges uniformly.

Example

$(x^n)_{n \in \mathbb{N}}$ does not converge uniformly on $[0, 1]$.

Proof idea

We show that it converges uniformly on $[0, c]$, for any $0 < c < 1$, and we highlight what part of the argument fails for $c=1$.

$(x^n)_{n \in \mathbb{N}}$ converges uniformly ^{to 0} on $[0, c]$, for $0 < c < 1$.

Let $\varepsilon > 0$. We want to find n_0 such that $|x^n - 0| < \varepsilon$ for $n \geq n_0$.

("Draft": $|x^n - 0| = x^n \leq c^n < \varepsilon \Leftrightarrow n = \frac{\ln(\varepsilon)}{\ln(c)}$. Since $x^n \leq x^{n_0}$ if $n \geq n_0$, we let $n_0 \geq \ln(\varepsilon) / \ln(c)$).

Let $n_0 \geq \ln(\varepsilon) / \ln(c)$. Then, for $n \geq n_0$,

$$|x^n - 0| = x^n \leq x^{n_0} \leq c^{n_0} \leq c^{\frac{\ln(\varepsilon)}{\ln(c)}} = \varepsilon.$$

So $(x^n)_{n \in \mathbb{N}}$ converges uniformly to 0 on $[0, c]$.

• Why it does not work when $c=1$?

If $c=1$, $\ln(c) = \ln(1) = 0$, so $\frac{\ln(\varepsilon)}{\ln(c)}$ is undefined.

As $c \rightarrow 1$, n_0 has to be larger and larger. 

Theorem

If $f_n \rightarrow F$ uniformly and f_n is continuous for all $n \in \mathbb{N}$, then F is continuous.

Proof next class...